

3 hours on examples

Problem 1 (20HMMTA4)

Using f as horseshoe ($\backslash\text{mho}$ is not available in my typeset thing).

$$\begin{aligned}
 \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{f(n,k)}{3^{n+k-7}} &= \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \sum_{\substack{\text{prime } p|n \\ p \geq k}} \frac{1}{3^{n+k-7}} \\
 &= \sum_{\text{prime } p} \sum_{p|n} \sum_{k=1}^p \frac{1}{3^{n+k-7}} \\
 &= \sum_{\text{prime } p} \sum_{p|n} \frac{3}{2} \left(\frac{1}{3^{n-6}} - \frac{1}{3^{p+n-6}} \right) \\
 &= \frac{3^7}{2} \sum_{\text{prime } p} \left(1 - \frac{1}{3^p} \right) \sum_{p|n} \frac{1}{3^n} \\
 &= \frac{3^7}{2} \sum_{\text{prime } p} \left(1 - \frac{1}{3^p} \right) \left(\frac{\frac{1}{3^p}}{1 - \frac{1}{3^p}} \right) \\
 &= \frac{3^7}{2} \sum_{\text{prime } p} \frac{1}{3^p} \\
 &= \frac{3^7}{2} \left(\frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^5} + \frac{1}{3^7} \right) + \frac{3^7}{2} \sum_{\text{prime } p \geq 11} \frac{1}{3^p} \\
 &= \frac{3^5 + 3^4 + 3^2 + 1}{2} + \frac{3^7}{2} \sum_{\text{prime } p \geq 11} \frac{1}{3^p} \\
 &= 167 + \frac{3^7}{2} \sum_{\text{prime } p \geq 11} \frac{1}{3^p}
 \end{aligned}$$

Note that $\frac{3^7}{2} \sum_{\text{prime } p \geq 11} \frac{1}{3^p} < \frac{3^7}{2} \sum_{p \geq 11} \frac{1}{3^p} = \frac{3}{2} \cdot \frac{3^{-4}}{2} < \frac{1}{2}$ so we are done.

Problem 4 (1802PFA)

$$\begin{aligned}
 &\int_0^1 \int_0^{\sqrt{x}} \frac{2xy}{1-y^4} dy dx \\
 &= \int_0^1 \int_{y^2}^1 \frac{2xy}{1-y^4} dx dy
 \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \frac{y}{1-y^4} \cdot (x^2) \Big|_{y^2}^1 dy \\
&= \int_0^1 y dy = \frac{1}{2}
\end{aligned}$$

5 clubs; 4 hours

Problem 5 (ZB63E600)

Note that,

$$\begin{aligned}
\frac{1}{n^3(n+1)^3} &= \frac{1}{n^3(n+1)^2} - \frac{1}{n^2(n+1)^3} \\
&= \left(\frac{1}{n^3(n+1)^1} - \frac{1}{n^2(n+1)^2} \right) - \left(\frac{1}{n^2(n+1)^2} - \frac{1}{n^1(n+1)^3} \right) = \dots
\end{aligned}$$

(Calculated this using some reverse Pascal's triangle idea)

$$= \frac{1}{n^3} - \frac{3}{n^2} + \frac{6}{n} - \frac{6}{n+1} - \frac{3}{(n+1)^2} - \frac{1}{(n+1)^3}$$

Therefore,

$$\begin{aligned}
&\sum_{n \geq 1} \frac{1}{n^3(n+1)^3} \\
&= \sum_{n \geq 1} \frac{1}{n^3} - \frac{1}{(n+1)^3} - 3 \sum_{n \geq 1} \frac{1}{n^2} + \frac{1}{(n+1)^2} + 6 \sum_{n \geq 1} \frac{1}{n} - \frac{1}{n+1} \\
&= 1 - 3 \left(2 \cdot \frac{\pi^2}{6} - 1 \right) + 6 \\
&= 10 - \pi^2
\end{aligned}$$

Problem 6 (10AMO5)

Note that,

$$\frac{1}{q(q+1)(q+2)} = \frac{\left(\frac{1}{2}\right)}{q} - \frac{1}{q+1} + \frac{\left(\frac{1}{2}\right)}{q+2}$$

Therefore,

$$2S_q = \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) + \left(\frac{1}{5} - \frac{2}{6} + \frac{1}{7} \right) + \dots + \left(\frac{1}{q} - \frac{2}{q+1} + \frac{1}{q+2} \right)$$

For $p = 3$, we have $\frac{1}{3} - \frac{2}{2 \cdot 3 \cdot 4} = \frac{1}{4}$. This clearly works.

For $p \neq 3$, we know $3 \nmid p$ meaning $+\frac{1}{p}$ is in the summation $2S_q$. Since $2p < q + 2$, all other terms in the summation have denominator relatively prime with p . Therefore, n is not divisible by p .

$$\begin{aligned} p &\mid m - n \\ \Leftrightarrow p &\mid \frac{m - n}{n} \\ \Leftrightarrow \frac{m}{n} &\equiv 1 \pmod{p} \end{aligned}$$

We have,

$$\begin{aligned} \frac{1}{p} - 2S_q &= \frac{m}{n} \\ &\equiv - \left(\left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{q+2} \right) - \frac{1}{p} - \left(\frac{3}{3} + \frac{3}{6} + \dots + \frac{3}{q+1} \right) \right) \\ &\equiv - \left(\left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{\left(\frac{3p-1}{2}\right)} \right) - \frac{1}{p} - \left(1 + \frac{1}{2} + \dots + \frac{1}{\left(\frac{p-1}{2}\right)} \right) \right) \\ &\equiv - \left(\left(\frac{1}{\left(\frac{p+1}{2}\right)} + \frac{1}{\left(\frac{p+3}{2}\right)} + \dots + \frac{1}{\left(\frac{3p-1}{2}\right)} \right) - 1 - \frac{1}{p} \right) \\ &\equiv - \left(\left(\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{p-1} \right) - 1 \right) \\ &\equiv 1 \pmod{p} \end{aligned}$$

10 clubs; 5.5 hours

Problem 7 (18099PS)

We first prove the magnitude of the expression is $\sqrt{p}$. Note that,

$$\begin{aligned} &\left(\sum_{x=0}^{p-1} \exp\left(\frac{2\pi i x^2}{p}\right) \right) \left(\sum_{x=0}^{p-1} \exp\left(-\frac{2\pi i x^2}{p}\right) \right) \\ &= \sum_{x=0}^{p-1} \sum_{y=0}^{p-1} \exp\left(\frac{2\pi i (x^2 - y^2)}{p}\right) \end{aligned}$$

$$= \sum_{x=0}^{p-1} \sum_{y=0}^{p-1} \exp\left(\frac{2\pi i(x-y)(x+y)}{p}\right)$$

We let $a = x - y$,

$$= \sum_{a \equiv 0 \pmod p} \sum_{y=0}^{p-1} \exp\left(\frac{2\pi i(a)(a+2y)}{p}\right)$$

For $a \equiv 0 \pmod p$, $\sum_{y=0}^{p-1} \exp\left(\frac{2\pi i(a)(a+2y)}{p}\right) = p$. Otherwise, $a(a+2y)$ would never repeat taken modulo p iterating $y = 0 \rightarrow p-1$ meaning $\sum_{y=0}^{p-1} \exp\left(\frac{2\pi i(a)(a+2y)}{p}\right) = 0$ for $a \not\equiv 0 \pmod p$. The sum is hence p meaning magnitude is $\sqrt{p}$ as desired.

Now note that $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$. This is 1 if and only if $p \equiv 1 \pmod 4$. If $p \equiv 1 \pmod 4$, then $\left(\frac{x}{p}\right) = 1 \Leftrightarrow \left(\frac{-x}{p}\right) = 1$. Since each may be present exactly twice, we may pair the two together to see that $\sum_{x=0}^{p-1} \exp\left(\frac{2\pi i x^2}{p}\right)$ is real.

On the other hand, if $p \equiv 3 \pmod 4$, then $\left(\frac{-1}{p}\right) = -1$ means every primitive root other than 1 appears twice in $\sum_{x=0}^{p-1} \exp\left(\frac{2\pi i x^2}{p}\right)$ and twice in $\overline{\sum_{x=0}^{p-1} \exp\left(\frac{2\pi i x^2}{p}\right)}$. We have 1 appears in once in each sum.

Summing it all up, $\sum_{x=0}^{p-1} \exp\left(\frac{2\pi i x^2}{p}\right) + \overline{\sum_{x=0}^{p-1} \exp\left(\frac{2\pi i x^2}{p}\right)} = 2 \sum_{a=0}^{p-1} e^{\frac{2\pi i a}{p}} = 0$ meaning this is completely complex as desired.

Problem 9 (15PTNMA3)

Lemma: If $\omega = \exp\left(\frac{2\pi i}{n}\right)$, then $\prod_{i=1}^n (1 + \omega^i) = 2$ for odd n .

Proof: Note that the polynomial $x^n + 1 = (x+1)(x+\omega)(x+\omega^2) \dots$. Plug in $x = 1$ to finish. $\square$

Note that,

$$\log_2 \left(\prod_{a=1}^{2015} \prod_{b=1}^{2015} \left(1 + e^{\frac{2\pi i ab}{2015}} \right) \right) = \sum_{a=1}^{2015} \log_2 \left(\prod_{b=1}^{2015} \left(1 + e^{\frac{2\pi i ab}{2015}} \right) \right)$$

We casework on $\gcd(a, 2015)$.

If we let $d = \gcd(a, 2015)$, we have $\omega^a, \omega^{2a}, \dots, \omega^{da}$ would be the $\frac{2015}{d}$ th roots of unity. This would repeat d times. In other words,

$$\log_2 \left(\prod_{b=1}^{2015} \left(1 + e^{\frac{2\pi i ab}{2015}} \right) \right) = \log_2 \left(\left(\prod_{b=1}^{\frac{2015}{d}} \left(1 + e^{\frac{2\pi i ab}{2015}} \right) \right)^d \right) = \log_2(2^d) = \gcd(a, 2015)$$

The answer is $\sum_{a=1}^{2015} \gcd(a, 2015)$. It is known that this is multiplicative function, so,

$$\begin{aligned} \sum_{a=1}^{2015} \gcd(a, 2015) &= \left(\sum_{a=1}^5 \gcd(a, 5) \right) \left(\sum_{a=1}^{13} \gcd(a, 13) \right) \left(\sum_{a=1}^{31} \gcd(a, 31) \right) \\ &= (5 + 1 \cdot 4)(13 + 1 \cdot 12)(31 + 1 \cdot 30) = 9 \cdot 25 \cdot 61 = 13725 \end{aligned}$$

18 clubs; 7.5 hours

Problem 10 (09ELMO6)

We show that, with quadratic reciprocity,

$$\begin{aligned} x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \frac{x^6}{6} + \dots &\equiv 0 \pmod{p} \\ \Leftrightarrow x - \frac{x^2}{2} + \frac{1}{3} - \frac{x}{4} + \frac{x^2}{5} - \frac{1}{6} + \dots &\equiv 0 \pmod{p} \end{aligned}$$

Case 1: $p = 3$

This can easily be checked to work.

Case 2: $p \equiv 1 \pmod{6}$

The sum goes to,

$$\left(\frac{1}{3} - \frac{1}{6} + \dots + \frac{1}{p-4} - \frac{1}{p-1} \right) + x \left(\frac{1}{1} - \frac{1}{4} + \dots - \frac{1}{p-3} \right) + x^2 \left(-\frac{1}{2} + \frac{1}{5} - \dots + \frac{1}{p-2} \right)$$

Note that $-\frac{1}{p-1} \equiv \frac{1}{1} \pmod{p}$; $\frac{1}{p-4} \equiv -\frac{1}{4} \pmod{p}$, ...

$$\equiv (1+x) \left(\frac{1}{1} - \frac{1}{4} + \dots - \frac{1}{p-3} \right) + x^2 \left(-\frac{1}{2} + \frac{1}{5} - \dots + \frac{1}{p-2} \right)$$

Since p divides $x^3 - 1$ but not $x - 1$, we have $p \mid x^2 + x + 1$,

$$\begin{aligned} &\equiv (-x^2) \left(\frac{1}{1} - \frac{1}{4} + \dots - \frac{1}{p-3} \right) + x^2 \left(-\frac{1}{2} + \frac{1}{5} - \dots + \frac{1}{p-2} \right) \\ &\equiv x^2 \left(-\frac{1}{1} - \frac{1}{2} + \frac{1}{4} + \frac{1}{5} - \dots + \frac{1}{p-3} + \frac{1}{p-2} \right) \pmod{p} \end{aligned}$$

We use the harmonic modulo p trick to get,

$$\begin{aligned}
& -\frac{1}{1} - \frac{1}{2} + \frac{1}{4} + \frac{1}{5} - \dots + \frac{1}{p-3} + \frac{1}{p-2} \\
& \equiv \frac{1}{p} \left(-\binom{p}{1} + \binom{p}{2} - \binom{p}{4} + \binom{p}{5} - \dots - \binom{p}{p-3} + \binom{p}{p-2} \right)
\end{aligned}$$

With generating function,

$$\frac{x(1+x)^p}{3} = \frac{\sum \binom{p}{k} x^{k+1}}{3}$$

Plugging in $x = 1, \exp\left(\frac{2\pi i}{3}\right), \exp\left(\frac{4\pi i}{3}\right)$ and summing gives,

$$\binom{p}{2} + \binom{p}{5} + \dots + \binom{p}{p-2} = \frac{2^p - 2}{3}$$

With generating function $\frac{x^2(1+x)^p}{3} = \frac{\sum \binom{p}{k} x^{k+2}}{3}$, we have,

$$\begin{aligned}
& \binom{p}{1} + \binom{p}{4} + \dots + \binom{p}{p} = \frac{2^p + 1}{3} \\
& \Leftrightarrow \binom{p}{1} + \binom{p}{4} + \dots + \binom{p}{p-3} = \binom{p}{2} + \binom{p}{5} + \dots + \binom{p}{p-2}
\end{aligned}$$

Thus,

$$\frac{1}{p} \left(-\binom{p}{1} + \binom{p}{2} - \binom{p}{4} + \binom{p}{5} - \dots - \binom{p}{p-3} + \binom{p}{p-2} \right) \equiv 0 \pmod{p}$$

Case 3: $p \equiv 5 \pmod{6}$.

The sum goes to,

$$\begin{aligned}
& \left(\frac{1}{3} - \frac{1}{6} + \dots + \frac{1}{p-2} \right) + x \left(\frac{1}{1} - \frac{1}{4} + \dots - \frac{1}{p-1} \right) + x^2 \left(-\frac{1}{2} + \frac{1}{5} - \dots - \frac{1}{p-3} \right) \\
& \equiv (1+x^2) \left(-\frac{1}{2} + \frac{1}{5} - \dots - \frac{1}{p-3} \right) + x \left(\frac{1}{1} - \frac{1}{4} + \dots - \frac{1}{p-1} \right) \\
& \equiv x \left(\frac{1}{1} + \frac{1}{2} - \frac{1}{4} - \frac{1}{5} + \dots + \frac{1}{p-4} + \frac{1}{p-3} - \frac{1}{p-1} \right)
\end{aligned}$$

Similar finish...

Problem 11 (95IMO6)

We solve for the number of subsets where p divides the sum and size. This gives us 2 extra subsets (empty and everything) which we will subtract from our result.

Define generating function,

$$F(x, y) = \prod_{k=1}^{2p} (1 + xy^k)$$

The x -degree of each term in the expansion corresponds to size while that of y corresponds to sum. We count the sum of coefficients of the terms with p dividing both degrees. Where ω is the p -th root of unity, this would be,

$$\frac{\sum_{a=1}^p \sum_{b=1}^p F(\omega^a, \omega^b)}{p^2}$$

We see that, if $b \neq p$,

$$\prod_{k=1}^{2p} (1 + \omega^{a+bk}) = \prod_{k=1}^{2p} (1 + \omega^k) = 2^2$$

If $b = p$,

$$\prod_{k=1}^{2p} (1 + \omega^{a+bk}) = (1 + \omega^a)^{2p}$$

The sum is therefore,

$$\frac{\sum_{a=1}^p \sum_{b=1}^p F(\omega^a, \omega^b)}{p^2} = \frac{4p(p-1) + \sum_{a=1}^p (1 + \omega^a)^{2p}}{p^2}$$

By reverse roots of unity filter, $\sum_{a=1}^p (1 + \omega^a)^{2p}$ is p times the sum of coefficients of the terms of $(1 + x)^{2p}$ where p divides the degree. Therefore,

$$\begin{aligned} &= \frac{4p(p-1) + p \left(\binom{2p}{0} + \binom{2p}{p} + \binom{2p}{2p} \right)}{p^2} \\ &= \frac{(4p-4) + \left(\binom{2p}{p} + 2 \right)}{p} \end{aligned}$$

The answer is,

$$\frac{(4p-4) + \left(\binom{2p}{p} + 2 \right)}{p} - 2 = \frac{\binom{2p}{p} + 2p - 2}{p}$$

28 clubs; 10.5 hours

Problem 13 (Z2E52572)

We have,

$$\begin{aligned} \sum_{x \in S} \frac{1}{x-1} &= \sum_{x \in S} \sum_{k \geq 1} \frac{1}{x^k} = \sum_{\substack{b \geq 2 \\ b \text{ not a power}}} \sum_{n \geq 2} \sum_{k \geq 1} \frac{1}{b^{nk}} \\ &= \sum_{\substack{b \geq 2 \\ b \text{ not a power}}} \sum_{k \geq 1} \sum_{n \geq 2} \frac{1}{b^{nk}} = \sum_{\substack{b \geq 2 \\ b \text{ not a power}}} \sum_{k \geq 1} \frac{1}{b^k (b^k - 1)} \end{aligned}$$

Note that the set $\{b^k \mid b \text{ not a power}, k \geq 2\}$ is the set of all positive integers ≥ 2 .

$$= \sum_{n \geq 2} \frac{1}{n(n-1)} = 1$$

Problem 16 (07SLC3)

Let $r_1, r_2, \dots, r_k$ be the red numbers and $b_1, b_2, \dots, b_{n-k}$ be the blue ones. We define generating functions,

$$R(x) = \sum_{i=1}^k x^{r_i}; B(x) = \sum_{i=1}^{n-k} x^{b_i}$$

Note that $R(x) + B(x) = x + x^2 + \dots + x^n$. What we want would be, after roots of unity filter,

$$\begin{aligned} &\frac{\sum_{i=1}^n R(\omega^i)^3 + B(\omega^i)^3}{n} \\ &= \frac{R(1)^3 + B(1)^3 + \sum_{i=1}^{n-1} R(\omega^i)^3 + B(\omega^i)^3}{n} \\ &= \frac{R(1)^3 + B(1)^3 + \sum_{i=1}^{n-1} (R(\omega^i) + B(\omega^i)) (R(\omega^i)^2 - R(\omega^i)B(\omega^i) + B(\omega^i)^2)}{n} \end{aligned}$$

Since $R(\omega^i) + B(\omega^i) = \frac{x^{n+1}-x}{x-1}$ is zero for $1 \leq i \leq n-1$,

$$\begin{aligned} &= \frac{R(1)^3 + B(1)^3}{n} \\ &= \frac{k^3 + (n-k)^3}{n} \\ &= (n-k)^2 - k(n-k) + k^2 \\ &= n^2 - 3nk + 3k^2 \end{aligned}$$

We look at $n^2 - 3nk + 3k^2 = 2007$.

$$k = \frac{3n \pm \sqrt{9n - 12(n^2 - 2007)}}{6}$$

Taking mod 3, $n^2 \equiv 0 \pmod 3$ meaning $3 \mid n$. We plug in $n = 3N$. Next, taking mod 9, we have $3k^2 \equiv 0 \pmod 9$. Therefore, $3 \mid k$ also and we write $k = 3K$.

$$\begin{aligned} N^2 - 3NK + 3K^2 &= 223 \\ 3K^2 + (-3N)K + (N^2 - 223) &= 0 \\ K &= \frac{3N \pm \sqrt{9N - 12(N^2 - 223)}}{6} \end{aligned}$$

We first make sure $9N - 12(N^2 - 223) = m^2$ is a square. Clearly, $3 \mid N$ now fails since $v_3(9N - 12(N^2 - 223)) = 1$. More casework on 1 mod 3 and 2 mod 3 may simplify further and gives answer of $n = 69, 84$.

36 clubs; 13.5 hours

Problem 18 (15AMO6)

We let real numbers $x_n := n\lambda - |A_n|$ be a sequence. FTSOC, if the sum of A_n is greater than $\frac{n(n+1)}{2}\lambda$ for all $n \geq k$,

$$\begin{aligned} S(A_n) &> \frac{n(n+1)}{2}\lambda \\ \Leftrightarrow 1(|A_1|) + 2(|A_2| - |A_1|) + 3(|A_3| - |A_2|) + \dots &> \frac{n(n+1)}{2}\lambda \\ \Leftrightarrow n|A_n| - |A_1| - |A_2| - \dots - |A_{n-1}| &> \frac{n(n+1)}{2}\lambda \\ \Leftrightarrow n(n\lambda - x_n) - (\lambda - x_1) - (2\lambda - x_2) - \dots - ((n-1)\lambda - x_{n-1}) &> \frac{n(n+1)}{2}\lambda \\ \Leftrightarrow \left(n^2 - \frac{(n-1)n}{2}\right)\lambda + (x_1 + \dots + x_{n-1}) - nx_n &> \frac{n(n+1)}{2}\lambda \\ \Leftrightarrow \frac{x_1 + \dots + x_{n-1}}{n} &> x_n \end{aligned}$$

Note that $x_n = n\lambda - |A_n| \geq 0$. Additionally, $x_{n+1} - x_n = \lambda - c$ for some non-negative integer c . Therefore, $|x_{n+1} - x_n| \geq \min(\lambda, 1 - \lambda)$. We let $\epsilon = \min(\lambda, 1 - \lambda)$. Note that $\epsilon > 0$ is fixed.

Since $x_n + x_{n+1} < \frac{x_1 + \dots + x_{n-1}}{n} + \frac{x_1 + \dots + x_{n-1}}{n} - \epsilon$ must be true,

$$\frac{x_1 + \dots + x_{n+1}}{n+2} < \frac{x_1 + \dots + x_{n-1}}{n} - \frac{\epsilon}{n+2}$$

We may then repeat this to get,

$$\frac{x_1 + \dots + x_{n+3}}{n+4} < \frac{x_1 + \dots + x_{n+1}}{n+2} - \frac{\epsilon}{n+4}$$

Inducting, we'd get that,

$$\frac{x_1 + \dots + x_{n+2k-1}}{n+2k} < \frac{x_1 + \dots + x_{n-1}}{n} - \frac{\epsilon}{n+2} - \dots - \frac{\epsilon}{n+2k}$$

Since the harmonic series does not converge, the RHS will dip below zero for large enough k which contradicts x being positive.

Problem 19 (10USATSTS8)

Define generating function,

$$\begin{aligned} F(m) &= \sum_{m \geq n} \left(\sum_{k=1}^n (-1)^{n-k} \binom{n}{k} k^m \right) x^m = \sum_{k=1}^n \sum_{m \geq n} (-1)^{n-k} \binom{n}{k} (xk)^m \\ &= \sum_{k=1}^n (-1)^{n-k} \binom{n}{k} \sum_{m \geq n} (xk)^m = \sum_{k=1}^n (-1)^{n-k} \binom{n}{k} \frac{(kx)^n}{1-kx} \end{aligned}$$

On the other hand, let,

$$\begin{aligned} G(m) &= \sum_{m \geq n} \left(\sum_{\substack{a_1 + \dots + a_n = m \\ a_i \geq 1}} 1^{a_1} 2^{a_2} \dots n^{a_n} \right) x^m = \sum_{a_i \geq 1} 1^{a_1} 2^{a_2} \dots n^{a_n} x^{a_1 + a_2 + \dots + a_n} \\ &= \sum_{a_i \geq 1} (x^{a_1}) (2x^{a_2}) \dots (nx^{a_n}) = \prod_{k=1}^n \frac{kx}{1-kx} \end{aligned}$$

We would finish if $F(m) = G(m)$ since all the coefficients would match up.

$$\begin{aligned} \prod_{k=1}^n \frac{kx}{1-kx} &= \sum_{k=1}^n (-1)^{n-k} \binom{n}{k} \frac{(kx)^n}{1-kx} \\ \Leftrightarrow \prod_{k=1}^n \frac{kx}{1-kx} &= \sum_{k=1}^n (-1)^{n-k} \binom{n}{k} \frac{(kx)^n}{1-kx} \end{aligned}$$

Factoring out x^n from both sides,

$$\Leftrightarrow \prod_{k=1}^n \frac{k}{1-kx} = \sum_{k=1}^n (-1)^{n-k} \binom{n}{k} \frac{k^n}{1-kx}$$

We get rid of the denominators, [Is doing this and then plugging in $x = 1, \dots, \frac{1}{n}$ mathematically correct or is there some divide-by-zero shenanigan?]

$$n! = \sum_{k=1}^n (-1)^{n-k} \binom{n}{k} k^n \prod_{\substack{j=1 \\ j \neq k}}^n (1 - jx)$$

The RHS is a degree $n - 1$ polynomial. Plugging in $x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}$, we show that the RHS is always $n!$ to finish.

We plug in $x = \frac{1}{a}$.

$$\sum_{k=1}^n (-1)^{n-k} \binom{n}{k} k^n \prod_{\substack{j=1 \\ j \neq k}}^n (1 - jx)$$

The inner product would be zero for $k \neq a$,

$$\begin{aligned} & (-1)^{n-a} \binom{n}{a} a^n \prod_{\substack{j=1 \\ j \neq a}}^n \left(1 - \frac{j}{a}\right) \\ &= (-1)^{n-a} \binom{n}{a} a \prod_{\substack{j=1 \\ j \neq a}}^n (a - j) \\ &= \binom{n}{a} a(a-1)(a-2) \dots (1)(-1)(-2) \dots (-(n-a))(-1)^{n-a} \\ &= \binom{n}{a} a! (n-a)! = n! \end{aligned}$$

50 clubs; 17 hours

Problem 20 (17HMMTC8)

Consider generating function,

$$\prod_{1 \leq i < j \leq 16} \left(\frac{1}{2} + \frac{x_1 x_2}{2} \right)$$

Every connection has a $\frac{1}{2}$ chance of happening. The degree of a certain term in the extension is its degree in the graph. For all the frogs to have degrees multiples of 4, we use the roots of unity filter. The answer is, where ω is the fourth root of unity,

$$\begin{aligned} & \frac{1}{4^{16}} \sum_{1 \leq a_1, a_2, \dots, a_{16} \leq 4} \prod_{1 \leq i < j \leq 16} \left(\frac{1}{2} + \frac{\omega^{a_i + a_j}}{2} \right) \\ &= \frac{1}{2^{152}} \sum_{1 \leq a_1, a_2, \dots, a_{16} \leq 4} \prod_{1 \leq i < j \leq 16} (1 + \omega^{a_i + a_j}) \end{aligned}$$

Note that if there are two $a_i = a_j = 1$, two $a_i = a_j = 3$, or a pair $a_i = 2; a_j = 4$, then the overall product $\prod_{1 \leq i < j \leq 16} (1 + \omega^{a_i + a_j})$ would have one factor of zero.

Case 1: 16 twos.

$$\prod_{1 \leq i < j \leq 16} (1 + \omega^{a_i + a_j}) = 2^{120}$$

Case 2: 15 twos.

We cannot have any fours so we either have one or three. There are 16 ways to place the one/three. In the former case,

$$16 \prod_{1 \leq i < j \leq 16} (1 + \omega^{a_i + a_j}) = 16 \cdot 2^{105} \cdot (1 + \omega^3)^{15}$$

In the latter case,

$$16 \prod_{1 \leq i < j \leq 16} (1 + \omega^{a_i + a_j}) = 16 \cdot 2^{105} \cdot (1 + \omega)^{15}$$

Note that,

$$\begin{aligned} 16(2^{105} \cdot (1 + \omega^3)^{15} + 2^{105} \cdot (1 + \omega)^{15}) &= 16 \left(2^{105} (1 + \omega^3)^{15} (1 + \omega^{15}) \right) = 2^{10} (1 + \omega^3)^{16} \\ &= 2^{109} (1 - i)^{16} = 2^{10} \cdot 2^8 = 2^{11} \end{aligned}$$

Case 3: 14 twos.

We cannot have any fours, so we have 1 one and 1 three. There are $\frac{16!}{14!} = 240$ ways for placement.

$$240 \prod_{1 \leq i < j \leq 16} (1 + \omega^{a_i + a_j}) = 240 \cdot 2^{91} \cdot (1 + \omega^3)^{14} \cdot (1 + \omega)^{14} \cdot 2 = 15 \cdot 2^{110}$$

Case 4: No twos.

We must have at least 14 fours. We find that this is symmetric so there is the same amount.

In conclusion, the answer is,

$$\begin{aligned} \frac{1}{2^{15}} (2(15 \cdot 2^{110} + 2^{117} + 2^{120})) \\ = \frac{1167}{2^{41}} \end{aligned}$$

55 clubs; 18 hours